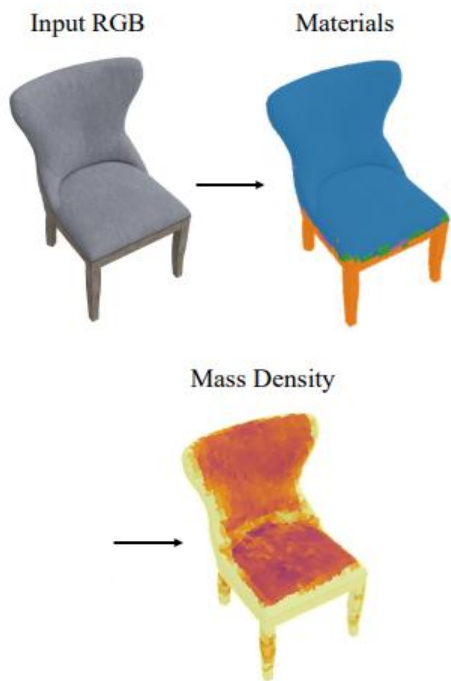


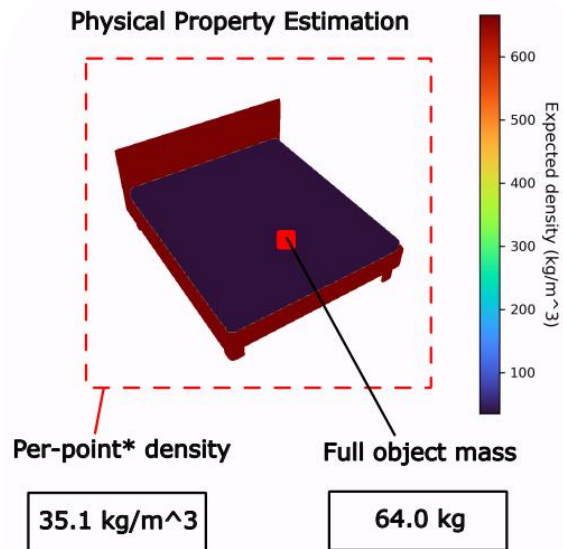
01 Physical Property Estimation

Infer material and mechanical properties of objects, including appearance, mass, rigidity and deformability, from visual observations.



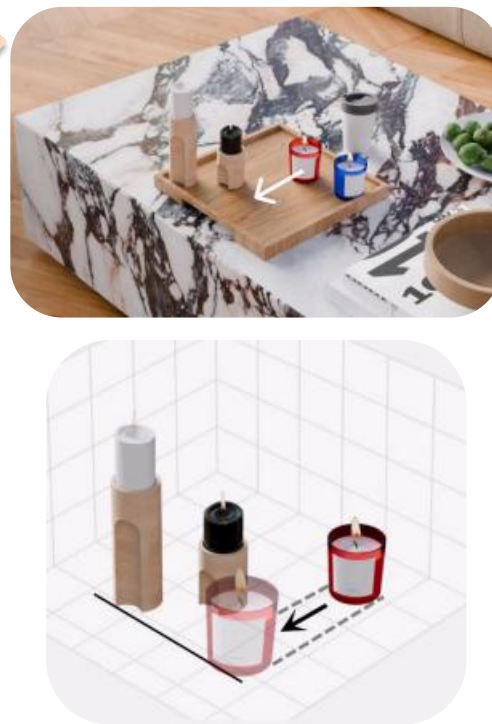
02 Affordance Reasoning

Reason actionable possibilities of objects, evolving from static geometry to human demonstration and active interactive learning.



03 Physical Causal Reasoning

Predict actions outcome, temporal dynamics and counterfactual physical events.



04 Physical Consistency Verification

Evaluate physical plausibility and validate executable simulation assets beyond visual realism.

Initial Frame



Later Frame

